Shuvomoy Das Gupta

Contact

6100 Main St MS-134, Houston, TX 77005, USA
https://shuvomoy.github.io/
sd158@rice.edu

Citizenship

Canada

Research Interests

Optimization, Game Theory, Computation, Transportation

Current Position

Rice University, Houston, TX, USA 2025-Present
Assistant Professor, Computational Applied Mathematics & Operations Research

Academic Experience

Columbia University, New York, NY, USA 2024-2025
Postdoctoral Research Scientist, Industrial Engineering & Operations Research
Hosts: Garud Iyengar & Christian Kroer
Worked on designing optimal algorithms for large-scale game solving.

Industry Experience

Thales Canada Inc., Toronto, Canada 2016-2018
Researcher, Research & Technology Department
Worked on real-time embedded optimization and sensor fusion algorithms in autonomous transportation systems.

Education

Massachusetts Institute of Technology, Cambridge, USA 2019 - 2024
Ph.D. in Operations Research
Thesis: Advances in Computer-Assisted Design and Analysis of First-Order Optimization Methods and Related Problems
PDF: https://dspace.mit.edu/handle/1721.1/155495
Advisors: Robert M. Freund & Bart P.G. Van Parys

University of Toronto, Toronto, Canada 2016
Master of Applied Science in Electrical and Computer Engineering
Thesis: Optimization Models for Energy-Efficient Railway Timetables
PDF: https://utoronto.scholaris.ca/items/b89b9f6b-f9db-4757-8971-6d1563f779fd
Advisor: Lacra Pavel

Grants, Awards and Honors

AFOSR Grant: "Computer-Assisted Design of Provably Fastest Algorithms". Co-PI. $600,000. (my share: $300,000) 2025-2028

Winner, INFORMS Computing Society Student Paper Award 2024

Honorable Mention & Finalist, INFORMS Nicholson Student Paper Competition 2024

Honorable Mention, MIT Operations Research Center Best Student Paper Award 2024

Preprints

[P1] Heechang Kim, Ernest K. Ryu, and Shuvomoy Das Gupta, "**A Domain-Specific Harness for End-to-End Automation of Optimization Research**," arXiv preprint arXiv:2608.07407, 2026.

[P2] Tianlong Nan, Garud Iyengar, Christian Kroer, and Shuvomoy Das Gupta, "**AltGDA Achieves Global $O(1/T)$ Ergodic Convergence in Matrix Games**," arXiv preprint arXiv:2609.32995, 2026.

[P3] Manu Upadhyaya, Shuvomoy Das Gupta, Adrien B. Taylor, Sebastian Banert, and Pontus Giselsson, "**The AutoLyap Software Suite for Computer-Assisted Lyapunov Analyses of First-Order Methods**," arXiv preprint arXiv:2506.24076, 2026.

[P4] Qiuyu Fan, J. Kevin Tobin, Bart P.G. Van Parys, and Shuvomoy Das Gupta, "**A Data-Driven Linear Programming Model for Energy-Optimal Metro Timetables**," arXiv preprint arXiv:2309.05489, 2026.

Published Journal Papers

[J1] Uijeong Jang, Shuvomoy Das Gupta, and Ernest K. Ryu, "**Computer-Assisted Design of Accelerated Composite Optimization Methods: OptISTA**," *Mathematical Programming*, 2026.

[J2] Shuvomoy Das Gupta, Bart P.G. Van Parys, and Ernest K. Ryu, "**Branch-and-Bound Performance Estimation Programming: A Unified Methodology for Constructing Optimal Optimization Methods**," *Mathematical Programming*, 2024.

[J3] Shuvomoy Das Gupta, Robert M. Freund, Xu Andy Sun, Adrien B. Taylor, "**Nonlinear Conjugate Gradient Methods: Worst-case Convergence Rates via Computer-assisted Analyses**," *Mathematical Programming*, 2024.

[J4] Shuvomoy Das Gupta, Bartolomeo Stellato, and Bart P.G. Van Parys, "**Exterior-point Optimization for Sparse and Low-rank Optimization**," *Journal of Optimization Theory and Applications*, 2024.

[J5] Shuvomoy Das Gupta and Lacra Pavel, "**On Seeking Efficient Pareto Optimal Points in Multi-Player Minimum Cost Flow Problems with Application to Transportation Systems**," *Journal of Global Optimization*, 2019.

[J6] Shuvomoy Das Gupta, J. Kevin Tobin, and Lacra Pavel, "**A Two-Step Linear Programming Model for Energy-Efficient Timetables in Metro Railway Networks**," *Transportation Research Part B: Methodological*, 2016.

Published Conference Papers

[C1] Tianlong Nan, Shuvomoy Das Gupta, Garud Iyengar, Christian Kroer, "**On the $O(1/T)$ Convergence of Alternating Gradient Descent–Ascent in Bilinear Games**," *International Conference on Learning Representations (ICLR)*, 2026.

[C2] Jakub Cerny, Shuvomoy Das Gupta, and Christian Kroer, "**Spatial Branch-and-Bound for Computing Multiplayer Nash Equilibrium**," *AAAI Conference on Artificial Intelligence (AAAI)*, 2026.

[C3] Shuvomoy Das Gupta, "**On Convergence of Heuristics Based on Douglas-Rachford Splitting and ADMM to Minimize Convex Functions over Nonconvex Sets**," *Allerton Conference on Communication, Control, and Computing*, 2018.

[C4] Shuvomoy Das Gupta and Lacra Pavel, "**Multi-player Minimum Cost Flow Problems with Nonconvex Costs and Integer Flows**," *IEEE Conference on Decision and Control (CDC)*, 2016.

[C5] Shuvomoy Das Gupta, Lacra Pavel, and J. Kevin Tobin, "**An Optimization Model to Utilize Regenerative Braking Energy in a Railway Network**," *American Control Conference (ACC)*, 2015.

Teaching

CMOR 537: Computer-assisted Algorithm Design for Optimization and Machine Learning, Rice Spring 2026
Instructor. New graduate course for its inaugural offering.

CMOR 467/567: Optimization for Energy Systems, Rice Fall 2025, 2026
Instructor. I have designed and launched this new course for advanced undergraduate and graduate students.

6.7220: Nonlinear Optimization, MIT Spring 2023
Teaching Assistant. This is MIT's main doctoral course in optimization.
Rating: 6.9/7.0

15.S60: Computing in Optimization and Statistics, MIT Winter 2023
Instructor. I taught the ORC's required three-hour module on advanced methods in computational optimization.
Rating: 7/7

15.S60: Computing in Optimization and Statistics, MIT Winter 2022
Instructor. I taught the ORC's required three-hour module on advanced methods in computational optimization.
Rating: 6.8/7

15.S08: Optimization of Energy Systems, MIT Spring 2022
Teaching Assistant. This is a graduate course in power systems modeling and optimization.
Rating: 6.0/7.0

Advising and Mentoring

Ph.D. students

- Qiuyu Fan (Rice University) 2025–

Undergraduate students

- Heechang Kim (Seoul National University) 2025–
- Uijeong Jang (Seoul National University; current: Ph.D. student at UCLA) 2022–23

Talks

A Domain-Specific Harness for End-to-End Automation of Optimization Research
INFORMS Annual Meeting, San Francisco, CA 2026
Decision Intelligence Talks Series, NVIDIA (virtual) 2026
MOPTA 2026, Lehigh University, Bethlehem, PA 2026

PEPit.jl: Computer-Assisted Worst-Case Analysis of First-Order Optimization Algorithms in Julia
JuMP-dev 2026, Edinburgh, Scotland 2026

On the $O(1/T)$ Convergence of Alternating Gradient Descent–Ascent in Bilinear Games
SIAM Conference on Optimization (OP26), Edinburgh, United Kingdom 2026
Mixed Integer Programming Workshop 2026, University of Connecticut - Stamford Campus 2026
INFORMS Annual Meeting, Atlanta, GA 2025

Computer-Assisted Design of Provably Fastest Algorithms
Invited talk, New Jersey Institute of Technology, New York, NY 2025

Nonlinear Conjugate Gradient Methods: Worst-case Convergence Rates via Computer-assisted Analyses
ICCOPT, Los Angeles, CA 2025
INFORMS Annual Meeting, Seattle, WA 2024

BnB-PEP: A Unified Methodology for Constructing Optimal Optimization Methods
INFORMS Annual Meeting, Phoenix, AZ 2023
SIAM Conference on Optimization (OP23), Seattle, Washington 2023
UTORG Seminar, University of Toronto, Toronto, Canada 2023
International Conference on Continuous Optimization, Bethlehem, PA 2022
MIT Data Science Lab Seminar 2022

Design and Analysis of First-Order Methods via Nonconvex QCQP Frameworks
One of just four invited "long talks" at the 1st Workshop on Performance Estimation, UCLouvain, Belgium 2023

Energy-Optimal Timetable Design for Sustainable Metro Railway Networks
INFORMS Annual Meeting, Phoenix, AZ 2023
33rd Annual POMS Conference, Orlando, FL 2023
2023 MIT Energy Initiative Annual Research Conference 2023

NExOS.jl for Nonconvex Exterior-point Optimization
JuliaCon 2021

Exterior-point Optimization for Sparse and Low-rank Optimization
Seoul National University 2020
INFORMS Annual Meeting (virtual) 2020
MIT LIDS student conference 2021

On Convergence of Heuristics Based on Douglas-Rachford Splitting and ADMM to Minimize Convex Functions over Nonconvex Sets
56th Allerton Conference on Communication, Control, and Computing, Monticello, IL 2018

Multi-Player Minimum Cost Flow Problems with Nonconvex Costs and Integer Flows
55th IEEE Conference on Decision and Control, Las Vegas, NV 2016

On seeking efficient Pareto optimal points in multi-player minimum cost flow problems
MIT LIDS student conference 2020

Service

Member, Mixed Integer Programming Workshop Program Committee 2027

Member, Land–Doig MIP Competition Organizing Committee 2027

Chair, Rice CMOR Colloquium Organization Committee 2026–2027
Coordinating the departmental colloquium program, including speaker invitations, scheduling, hosts, and seminar logistics.

Member, INFORMS George Nicholson Student Paper Competition 2026–2027

Member, Rice CMOR Undergraduate Committee 2026–2027
Supporting undergraduate curriculum, advising, major/minor requirements, and student-facing program resources.

Member, Rice CMOR Graduate Committee 2025–2026
Supported graduate program administration, including student advising, degree requirements, qualifying exams, and academic progress.

Organizer, Friends of Optimization Seminar Series 2024–
LINK: `https://sites.google.com/view/friends-of-optimization`

Reviewer for *Mathematical Programming, Management Science, Transportation Research Part B: Methodological, IEEE Transactions on Control of Network Systems, American Control Conference, IEEE Transactions on Intelligent Transportation Systems, IEEE Transactions on Automatic Control*

Session Chair, INFORMS Annual Meeting 2022–2025

Software

[S1] **AltGDA-code**
Julia experiments and Lean 4 formalization for the global ergodic convergence of AltGDA in matrix games.
LINK: `https://github.com/Shuvomoy/AltGDA-code`

[S2] **AutoOPT**
Domain-specific harness for end-to-end automation of optimization research through numerical algorithm design, symbolic discovery, and Lean verification.
LINK: `https://github.com/Shuvomoy/AutoOPT`

[S3] **PEPit.jl**
Native Julia implementation of the Performance Estimation Programming (PEP) methodology.
LINK: `https://github.com/PerformanceEstimation/PEPit.jl`

[S4] **AutoLyap.jl**
Native Julia implementation of the AutoLyap methodology.
LINK: `https://github.com/AutoLyap/AutoLyap.jl`

[S5] **BnB-PEP**
Computes optimal first-order algorithms for different convex and nonconvex setups
LINK: `https://github.com/Shuvomoy/BnB-PEP-code`

[S6] **NCG-PEP**
Computes worst-case convergence rates of nonlinear conjugate gradient methods
LINK: `https://github.com/Shuvomoy/NCG-PEP-code`

[S7] **NExOS**
Implements the Nonconvex Exterior-point Optimization Solver (NExOS) algorithm for solving low-rank and sparse optimization problems
LINK: `https://github.com/Shuvomoy/NExOS.jl`

Languages

Fluent in
English, Bengali, Hindi, Urdu
Proficient in
`Julia, C, C++, MATLAB, Mathematica`

Other

I enjoy playing cricket, reading novels, cooking, and blogging at `https://shuvomoy.github.io/blogs/`.

Media Coverage

"Risky Giant Steps Can Solve Optimization Problems Faster" August, 2023
by Allison Parshall in *Quanta Magazine*
I was interviewed and quoted in the article along with my paper being cited as the main inspiration for the discovery of long step gradient descent by Ben Grimmer. Also publicized in the *Nautilus Quarterly Magazine* and in the Chinese magazine *Heart of the Machine*.
URL: `https://www.quantamagazine.org/risky-giant-steps-can-solve-optimization-problems-faster-20230811/`